

CO3 AT2 OUTPUTS

1.

```
er' '63174' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO3 AT2\Q1.PY'
P(science | data) = 1.0
P(improves | data science) = 0
Interpolated probability = 0.5667
Entropy = 0.9235
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING> █
```

2.

```
er' '52756' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO3 AT2\Q2.PY'
Book -> VB
a -> DT
flight -> NN
ticket -> NN
now -> RB

Sentence 2:
This -> DT
book -> NN
is -> VBZ
interesting -> JJ

P(Book = VB) = 0.3
P(Book = NN) = 0.2
HMM chooses VB
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING> █
```

3.

```
er 51780 -- C:\Users\Freddy\hip\DSA0304-NATURAL-LANGUAGE-PROCE
Before Transformation:
economic -> JJ
growth -> NN
increases -> NNS
employment -> NN

After Transformation:
economic -> JJ
growth -> NN
increases -> VBZ
employment -> NN

Word Frequencies:
economic -> 120
growth -> 450
increases -> 210
employment -> 380

Total = 1160

Word Probabilities:
economic -> 0.103
growth -> 0.388
increases -> 0.181
```